



Software Requirements Specification (SRS)

CSC-225 – Software Engineering
Department of Computer Science
Namal University, Mianwali

Project Title: Namal Criko
Cricket Simulation Trainer

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1. Introduction

1.1. Purpose

1.1.1 Overview of the Document

This Software Requirements Specification (SRS) document provides a comprehensive description of the Namal Criko cricket simulation training platform. The document serves as the primary reference for all stakeholders involved in the development, testing, and deployment phases of the project. It establishes a clear understanding of the system's functionalities, constraints, and quality attributes, ensuring alignment between technical teams and the requirement provider.

1.1.2 Purpose of the SRS

The purpose of this SRS document is to:

- Define all functional and non-functional requirements for the Namal Criko platform in detail
- Serve as a baseline for project planning, design, implementation, and validation activities
- Provide clear specifications for quality assurance teams to develop comprehensive test cases
- Enable future maintenance teams to understand system architecture and design decisions
- Document assumptions, dependencies, and constraints that influence system development

1.1.3 Intended Audience

This document addresses the needs of multiple stakeholder groups:

Development Team

- **Software Engineers:** To implement features according to specified requirements
- **System Architects:** To design scalable and maintainable system architecture
- **UI/UX Designers:** To create intuitive interfaces aligned with user needs
- **Database Administrators:** To design efficient data storage and retrieval mechanisms

Quality Assurance Team

- **QA Engineers:** To develop test cases and validation strategies
- **Testing Specialists:** To ensure system meets all specified requirements

Project Management

- **Project Managers:** To plan resources, timelines, and deliverables
- **Requirement Provider (Mr. Ammar Ahmad Khan):** To validate that requirements are correctly captured

End Users

- **Cricket Students:** Individuals seeking to improve their cricket skills through simulation
- **Cricket Coaches:** Professionals using the platform as a training supplement
- **Administrators:** Staff members managing user accounts and content

1.2. Scope

1.2.1 Product Identification

Namal Criko is a web-based interactive cricket simulation training platform designed specifically for students and cricket enthusiasts at Namal University . The system provides a controlled, technology-driven environment where users can learn and practice various cricket techniques through visual simulations combined with comprehensive textual instruction.

1.2.2 Product Features

What Namal Criko Will Do:

- Provide interactive simulations for batting techniques including drives, sweeps, cuts, pulls, and defensive shots
- Offer bowling simulations covering fast bowling, spin bowling, and variations (Yorker, bouncer, slower ball)
- Deliver step-by-step visual guidance with body movement demonstrations and timing indicators
- Enable users to select specific techniques for focused practice sessions

- Track individual player performance metrics including matches played, run rate, and skill progression
- Implement hybrid learning through combination of interactive simulations and descriptive technical text
- Support administrative control over user access, content management, and performance monitoring
- Provide a help desk system for user query resolution and technical support
- Allow administrators to post cricket-related event announcements and advertisements

What Namal Criko Will Not Do (Initial Release):

- Provide real-time multiplayer cricket matches or competitive gameplay
- Include artificial intelligence-based opponent players
- Support virtual reality (VR) or augmented reality (AR) interfaces
- Offer mobile native applications (initial release is web-only)
- Integrate with external cricket statistics databases or APIs
- Provide live video streaming or recorded video tutorials
- Include payment gateways for premium features or subscriptions
- Support multiple languages (English only in initial release)

1.2.3 Application of the Software

Namal Criko addresses critical gaps in cricket training by providing:

- **Accessible Training:** Eliminates geographical and temporal constraints by providing 24/7 access to cricket training simulations
- **Cost-Effective Learning:** Reduces dependency on expensive coaching sessions and physical training facilities
- **Technique Mastery:** Enables focused practice on specific techniques with immediate visual feedback
- **Performance Tracking:** Provides data-driven insights into skill progression and areas requiring improvement

- **Safe Practice Environment:** Allows beginners to learn fundamental techniques without risk of injury
- **Supplementary Training Tool:** Complements physical practice by reinforcing theoretical understanding

1.2.4 Goals and Benefits

- **Skill Development:** Accelerate the learning curve for cricket techniques through interactive practice
- **Accessibility:** Provide cricket training resources to students who lack access to professional coaching
- **Engagement:** Increase student interest in cricket through gamified learning experiences
- **Data Analytics:** Enable coaches and administrators to monitor student progress systematically
- **Community Building:** Foster a cricket culture within Namal University through shared learning platform
- **Scalability:** Build foundation for future enhancements including multiplayer modes and advanced analytics

1.3. Overview

1.3.1 Document Structure

This SRS document follows the IEEE 830 standard and is organized into the following major sections:

1. **Introduction:** Provides document purpose, scope, definitions, and overview
2. **Overall Description:** Describes product perspective, user characteristics, constraints, and assumptions
3. **Specific Requirements:** Details functional requirements, interface specifications, and system features
4. **System Features:** Enumerates core features with detailed descriptions
5. **External Interface Requirements:** Specifies user, hardware, software, and communication interfaces

6. **Non-Functional Requirements:** Covers performance, security, reliability, and usability requirements
7. **Other Requirements:** Addresses legal, regulatory, and operational considerations
8. **Appendices:** Contains supplementary materials including diagrams, glossary, and references

1.3.2 Organization of the Document

Each section builds upon previous sections to provide progressively detailed specifications:

- Sections are numbered hierarchically for easy reference and navigation
- Cross-references link related requirements and specifications
- Technical terms are defined in the glossary to ensure consistent understanding
- Diagrams and tables supplement textual descriptions for clarity
- Requirements are uniquely identified with tags (e.g., FR-001, NFR-001) for traceability

1.4. Definitions, Acronyms, and Abbreviations

Term/Acronym	Definition
Namal Criko	The name of the cricket simulation training platform developed for Namal University students
SRS	Software Requirements Specification - a document describing system requirements
RP	Requirement Provider - Mr. Ammar Ahmad Khan, the stakeholder defining system requirements
UI/UX	User Interface/User Experience - design aspects affecting user interaction with the system
Admin	Administrator - privileged user with system management capabilities
Player	Regular user who accesses cricket training simulations
Simulation	Interactive visual representation of cricket techniques with step-by-step guidance
Run Rate	Performance metric measuring runs scored per over or per match
IEEE 830	IEEE Recommended Practice for Software Requirements Specifications standard

Term/Acronym	Definition
HTTPS	Hypertext Transfer Protocol Secure - encrypted communication protocol
DBMS	Database Management System - software for managing structured data
HTML5	HyperText Markup Language version 5 - standard for web content structure
CSS3	Cascading Style Sheets version 3 - stylesheet language for web presentation
JavaScript	Programming language for interactive web functionality
Responsive Design	Design approach ensuring optimal display across various device screen sizes
Session Management	System mechanism for tracking authenticated user interactions
Two-Factor Authentication (2FA)	Security mechanism requiring two forms of identity verification
CRUD Operations	Create, Read, Update, Delete - fundamental data manipulation operations
MVP	Minimum Viable Product - initial version with core features

1.5. References

1.5.1 Standards

1. IEEE Std 830-1998: IEEE Recommended Practice for Software Requirements Specifications
2. IEEE Std 1012-2016: IEEE Standard for System, Software, and Hardware Verification and Validation

1.5.2 Technical Documentation

- (a) HTML5 Specification: <https://html.spec.whatwg.org/>
- (b) CSS3 Standards: <https://www.w3.org/Style/CSS/>
- (c) JavaScript ES6+ Documentation: <https://developer.mozilla.org/en-US/docs/Web/JavaScript>
- (d) React Documentation (if applicable): <https://react.dev/>
- (e) Node.js Documentation (if applicable): <https://nodejs.org/docs/>
- (f) MongoDB Documentation (if applicable): <https://www.mongodb.com/docs/>

1.5.3 Project Documentation

- (a) Namal Criko Project Proposal (Milestone 1)
- (b) Meeting Minutes with Requirement Provider
- (c) Requirement Provider Agreement
- (d) Project GitHub Repository: <https://github.com/MJamalAhmadKhan/Namal-Criko>

1.5.4 Development Tools

- (a) Visual Studio Code: <https://code.visualstudio.com>
- (b) Git Version Control: <https://git-scm.com>
- (c) GitHub Documentation: <https://docs.github.com>

2. Overall Description

2.1. Product Perspective

Namal Criko is an independent web-based cricket training platform designed to supplement traditional physical training methods. Unlike commercial cricket games focused on entertainment, Namal Criko focuses educational content and technique mastery through interactive simulations paired with instructional materials.

2.1.1 System Context

The system operates as a standalone platform with the following contextual relationships:

Relationship with Physical Training Namal Criko serves as a complementary tool to physical cricket practice rather than a replacement. It provides:

- Theoretical understanding of techniques before physical practice
- Reinforcement of concepts learned during coaching sessions
- Convenient access to technique review between practice sessions
- Visual aids for self-correction of form and timing

Integration Possibilities While initially standalone, the architecture allows future integration with:

- University student information systems for automated user provisioning
- Cricket coaching management platforms for progress tracking
- Video analysis tools for comparing simulations with real performances
- Wearable fitness trackers for incorporating physical activity data

2.1.2 System Interfaces

The system comprises several interconnected components:

User Interface Layer

- Web browser-based interface supporting modern browsers (Chrome, Firefox, Safari, Edge)
- Responsive design adapting to desktop, tablet, and mobile screen sizes
- Intuitive navigation structure for quick access to simulations

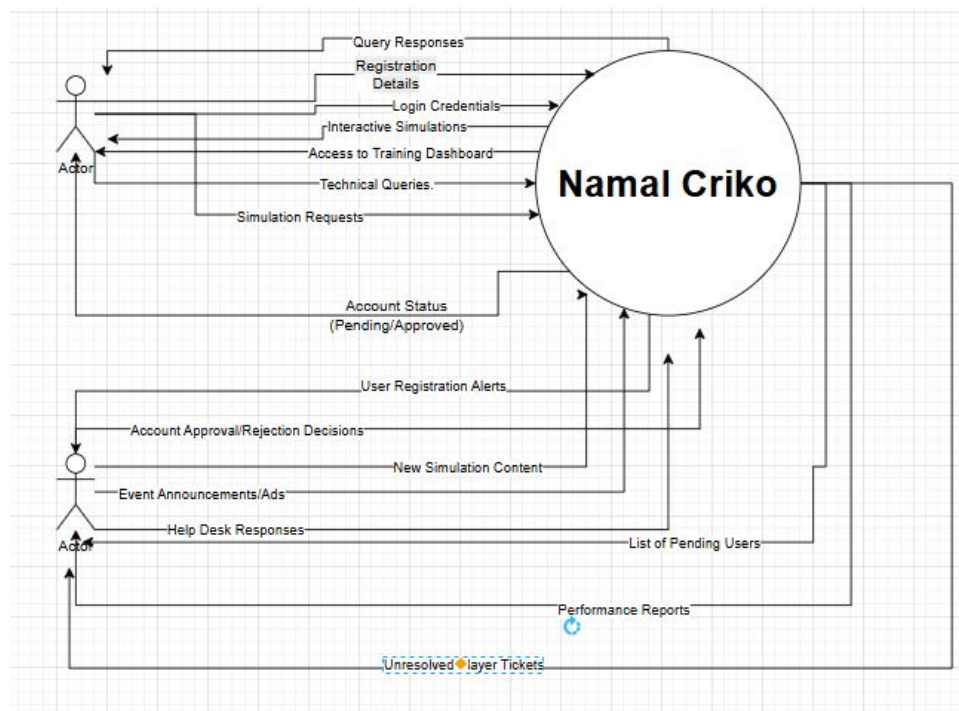


Figure 1: System Context Diagram showing external entities and data flows

Application Layer

- Business logic for user authentication and authorization
- Simulation engine managing technique demonstrations
- Performance tracking and analytics module
- Content management system for administrators

Data Layer

- Relational or NoSQL database storing user profiles, performance data, and content
- File storage system for simulation assets (images, animations)
- Session management for maintaining user state

2.1.3 User Classes and Characteristics

The system defines two primary user classes with distinct characteristics and access levels:

Player (Regular User)

- **Primary Goal:** Learn and practice cricket techniques

- **Technical Proficiency:** Basic to intermediate computer literacy
- **Age Range:** Primarily 15-25 years (university students)
- **Usage Frequency:** Expected 2-5 sessions per week, 15-30 minutes per session
- **Cricket Experience:** Ranges from complete beginners to intermediate players
- **Educational Background:** University-level education
- **Access Requirements:** Account creation subject to admin approval
- **Key Features Used:** Simulations, performance tracking, help desk

Administrator

- **Primary Goal:** Manage platform content, users, and system operations
- **Technical Proficiency:** Intermediate to advanced computer literacy
- **Responsibilities:** User account approval, content updates, query resolution, event management
- **Usage Pattern:** Daily monitoring with periodic content updates
- **Cricket Knowledge:** Strong understanding of cricket techniques and terminology
- **Access Requirements:** Pre-configured credentials with elevated privileges
- **Key Features Used:** User management, content management, analytics dashboard, help desk management

2.2. Product Functions

The core functions of Namal Criko are organized into three major functional areas:

2.2.1 Cricket Technique Simulations

Batting Simulations The system provides comprehensive batting technique simulations organized by shot type:

- **Defensive Techniques**
 - Forward Defense: Body position, bat angle, weight transfer
 - Backward Defense: Footwork, bat-pad gap, balance maintenance
 - Leaving Deliveries: Judgment, shoulder drops, bat withdrawal
- **Attacking Shots**

- Straight Drive: Timing, follow-through, elbow position
- Cover Drive: Footwork, bat path, weight transfer
- Square Drive: Bat angle, foot placement, shot control
- Pull Shot: Weight distribution, head position, shot timing
- Hook Shot: Technique variation, risk management, shot selection
- Cut Shot: Backward movement, bat angle, wrist work
- Sweep Shot: Positioning, bat angle, shot execution
- Reverse Sweep: Grip change, body position, shot control

Each simulation includes:

- Frame-by-frame breakdown of body movements
- Timing indicators for optimal shot execution
- Common mistakes and correction tips
- Variation demonstrations (e.g., different ball speeds, lengths)

Bowling Simulations The system covers major bowling categories with technique-specific guidance:

- **Fast Bowling**

- Run-up mechanics: Rhythm, acceleration, approach angle
- Delivery stride: Front foot placement, back foot drive, balance
- Arm action: Bowling arm path, release point, follow-through
- Variations: Yorker, bouncer, slower ball, knuckle ball

- **Spin Bowling**

- Off-Spin: Finger position, wrist action, loop trajectory
- Leg-Spin: Wrist position, ball release, turn generation
- Arm Ball: Grip variation, release modification
- Doosra/Googly: Advanced technique demonstrations

- **Medium-Pace Bowling**

- Swing bowling: Wrist position, seam orientation, release point
- Cutters: Grip variation, finger work, seam manipulation
- Change of pace: Rhythm variation, deception techniques

2.2.2 User Management and Progress Tracking

Player Profile Management

- User registration with validation (name, email, contact number)
- Profile customization options (preferred role, skill level, goals)
- Personal dashboard displaying recent activity and statistics
- Account status monitoring (pending approval, active, suspended)

Performance Analytics

- Technique completion tracking (which simulations completed)
- Time spent per technique category
- Progress visualization through charts and graphs
- Performance metrics including:
 - Total simulations accessed
 - Techniques mastered (based on completion frequency)
 - Learning streaks and consistency metrics
 - Comparative analysis against peer averages (optional)

Administrative User Management

- Account approval workflow with notification system
- User list management with search and filter capabilities
- Bulk operations for user status changes
- Activity logs for audit and compliance

2.2.3 Content and Communication Management

Simulation Content Management

- Upload and organize simulation assets (images, animations, text)
- Update existing simulation content without system downtime
- Version control for content iterations
- Content scheduling for timed releases

Help Desk System

- User-submitted query/ticket creation
- Category-based ticket classification (technical, content, account)
- Ticket status tracking (open, in-progress, resolved, closed)
- Admin response interface with email notifications
- FAQ management for common queries

Event and Announcement Management

- Cricket event posting with details (date, venue, registration info)
- Advertisement management for cricket-related promotions
- Notification system for upcoming events
- Event archiving and history maintenance

2.3. User Characteristics

2.3.1 Educational Level and Background

Namal Criko targets users with diverse educational backgrounds while maintaining accessibility:

Primary User Base

- **University Students:** Enrolled at Namal University or similar institutions
- **Education Level:** Undergraduate or equivalent qualification
- **Academic Disciplines:** All fields (not limited to sports sciences)
- **Computer Literacy:** Basic proficiency with web browsers and digital interfaces

Content Accessibility

- Instructions presented in clear, non-technical English
- Cricket terminology explained through tooltips and glossary
- Visual demonstrations complement textual descriptions
- Progressive difficulty levels accommodate beginners through advanced users

2.3.2 Cricket Experience Levels

The platform accommodates users across the cricket proficiency spectrum:

Beginner Level

- Users with minimal or no cricket experience
- Require foundational technique explanations
- Benefit from slow-motion breakdowns and detailed annotations
- Need extensive textual guidance alongside visual simulations

Intermediate Level

- Users with basic cricket knowledge from school or club participation
- Seek technique refinement and advanced variations
- Can follow standard-pace demonstrations with supplementary notes
- Interested in comparative analysis of different approaches

Advanced Level

- Users with competitive cricket experience
- Focus on fine-tuning specific aspects of technique
- Utilize platform for review and mental preparation
- May contribute feedback for content improvement

2.3.3 Technical Proficiency

Player Technical Requirements

- Ability to navigate web browsers and fill online forms
- Understanding of basic web interactions (clicking, scrolling, form submission)
- No programming or advanced technical knowledge required
- Familiarity with email for account notifications

Administrator Technical Requirements

- Intermediate computer skills including file management
- Understanding of content management principles
- Ability to review and analyze user data
- Basic troubleshooting skills for user support

2.3.4 Accessibility Considerations

The platform incorporates inclusive design principles:

- High contrast text for visual clarity
- Keyboard navigation support for users with mobility constraints
- Text alternatives for visual content where applicable
- Adjustable playback speeds for simulations
- Clear language avoiding unnecessary jargon

2.4. Constraints

2.4.1 Regulatory and Policy Constraints

Data Protection and Privacy

- Compliance with university data protection policies
- User consent requirements for data collection and processing
- Secure storage of personal information (names, emails, contact numbers)
- Data retention and deletion policies aligned with legal requirements
- Privacy policy clearly communicated to users during registration

Educational Use Policy

- Platform intended exclusively for educational purposes
- No commercial use or monetization of platform content
- Intellectual property rights maintained by Namal University
- User-generated content subject to university acceptable use policies

2.4.2 Hardware Limitations

Client-Side Constraints

- **Device Compatibility:** Must function on devices available to Namal University students (personal laptops, desktops, smartphones, tablets)
- **Internet Connectivity:** Performance dependent on user's internet connection quality; minimum 1 Mbps recommended
- **Screen Resolution:** Simulations optimized for minimum 1024x768 resolution but adapt to mobile devices

- **Browser Requirements:** Latest versions of Chrome, Firefox, Safari, or Edge; older browsers may have degraded functionality

Server-Side Constraints

- Hosting infrastructure limited by university IT budget
- Server capacity to handle peak concurrent users (estimated 100-200 simultaneous users)
- Storage constraints for simulation assets and user data
- Bandwidth limitations affecting media delivery speeds

2.4.3 Software Interface Constraints

Technology Stack Limitations

- Development constrained to open-source technologies to minimize licensing costs
- Framework selection influenced by team expertise and available documentation
- Database choice limited by hosting environment capabilities
- No integration with proprietary third-party cricket analytics platforms

Authentication Constraints

- Single sign-on (SSO) integration with university systems not feasible in initial release
- Email verification dependent on university email server reliability
- Password recovery mechanisms limited to email-based reset

2.4.4 Design and Implementation Constraints

Development Timeline

- Project must be completed within 1 year
- Milestone deadlines impose incremental delivery requirements
- Testing and refinement time limited by project schedule

Team Constraints

- Development team of three members with limited prior experience
- Part-time development due to concurrent academic commitments
- Knowledge limitations may necessitate simpler implementation approaches

Budget Constraints

- Zero-budget project restricting use of paid services or APIs
- Hosting on free-tier cloud services with associated limitations
- No budget for professional design assets or stock media

2.4.5 Operational Constraints**Maintenance and Support**

- Limited ongoing maintenance capacity post-semester
- Administrative responsibilities delegated to designated university staff
- Technical support dependent on administrator availability

Content Updates

- New simulation content creation requires administrator technical skills
- Content review and quality assurance time-consuming
- Updates may cause temporary service interruptions

2.5. Assumptions and Dependencies**2.5.1 Assumptions****User Behavior Assumptions**

- Users will access the platform with genuine intent to learn cricket techniques
- Users possess basic understanding of cricket rules and terminology
- Users will provide accurate personal information during registration
- Users will utilize the help desk appropriately without spam submissions
- Users have access to stable internet connections during learning sessions

Technical Assumptions

- Users' devices meet minimum hardware specifications for web browsing
- Modern web browsers with JavaScript enabled are standard among target users
- University network infrastructure provides adequate bandwidth
- Hosting platform maintains 99% uptime reliability
- Database systems handle concurrent access without significant performance degradation

Operational Assumptions

- Administrative staff available for user approval within 24-48 hours of registration
- Content experts available to validate cricket technique accuracy
- University continues to support project post-development
- User base grows gradually allowing for capacity planning

2.5.2 Dependencies

External Dependencies

- **Hosting Services:** Availability and performance of cloud hosting provider
- **Email Services:** Reliability of email delivery for notifications and verification
- **Browser Vendors:** Continued support for web standards and APIs used by the platform
- **Internet Service Providers:** User access dependent on ISP connectivity quality

Internal Dependencies

- **Development Team Availability:** Project progress dependent on team member schedules
- **Requirement Provider Input:** Timely feedback from Mr. Ammar Ahmad Khan essential for validation
- **University Resources:** Access to university servers or approved hosting alternatives
- **Content Creation:** Simulation quality dependent on availability of cricket technique references

Third-Party Dependencies

- Open-source frameworks and libraries for development
- Free-tier hosting services within usage limits
- Public cricket technique resources for content validation
- Development tools and IDEs for coding and testing

3. Specific Requirements

This section provides detailed specifications for all functional and interface requirements of the Namal Criko system. Each requirement is uniquely identified and described with sufficient detail for design and implementation.

3.1. Functional Requirements

3.1.1 User Authentication and Authorization

FR-001: User Registration

- **Description:** The system shall allow new users to create player accounts through a registration form
- **Inputs:** Full name, email address, contact number, password, confirm password
- **Process:**
 - (a) User accesses registration page
 - (b) User fills mandatory fields with validation checks
 - (c) System validates email format and password strength
 - (d) System creates account in "Pending Approval" status
 - (e) System sends confirmation email to user
 - (f) Admin receives notification of new registration
- **Outputs:** Account created with pending status, email notifications sent
- **Validation Rules:**
 - Email must be unique in system
 - Password minimum 8 characters with alphanumeric combination
 - Contact number must be valid format
 - All fields mandatory
- **Priority:** High

FR-002: Admin Account Approval

- **Description:** The system shall require administrator to manually approve new user accounts before granting access
- **Inputs:** Admin credentials, user account selection, approval/rejection decision
- **Process:**

- (a) Admin logs into admin portal
- (b) Admin views list of pending account approvals
- (c) Admin reviews user information
- (d) Admin approves or rejects account
- (e) System updates account status
- (f) System sends status notification email to user
- **Outputs:** Account status changed to "Active" or "Rejected", notification email sent
- **Business Rules:** Only approved accounts can access dashboard and simulations
- **Priority:** High

FR-003: User Login (Player)

- **Description:** The system shall authenticate players using email and password credentials
- **Inputs:** Email address, password
- **Process:**
 - (a) User enters credentials on player login page
 - (b) System verifies credentials against database
 - (c) System checks account approval status
 - (d) If approved, system creates user session and redirects to player dashboard
 - (e) If not approved, system displays "Account pending approval" message
 - (f) If credentials invalid, system displays error message
- **Outputs:** Authenticated session created, dashboard displayed
- **Security:** Failed login attempts tracked, account locked after 5 consecutive failures
- **Priority:** High

FR-004: Admin Login

- **Description:** The system shall provide separate authentication portal for administrators
- **Inputs:** Admin email, admin password
- **Process:** Similar to player login with redirect to admin dashboard
- **Outputs:** Authenticated admin session, admin dashboard displayed
- **Security:** Enhanced security with IP restrictions and session timeout
- **Priority:** High

FR-005: Password Reset

- **Description:** The system shall allow users to reset forgotten passwords via email verification
- **Inputs:** Email address for reset request
- **Process:**
 - (a) User clicks "Forgot Password" link
 - (b) User enters registered email address
 - (c) System generates unique password reset token
 - (d) System sends reset link to email
 - (e) User clicks link and enters new password
 - (f) System validates token and updates password
- **Outputs:** Password updated, confirmation message displayed
- **Security:** Reset tokens expire after 1 hour
- **Priority:** Medium

FR-006: Session Management

- **Description:** The system shall maintain secure user sessions with automatic timeout
- **Process:**
 - Active sessions valid for 120 minutes of inactivity
 - Warning displayed 5 minutes before timeout
 - User can extend session through activity
 - Logout functionality to explicitly terminate session
- **Priority:** Medium

3.1.2 Player Module Requirements**FR-007: Player Dashboard**

- **Description:** The system shall provide personalized dashboard displaying user profile and activity summary
- **Components:**
 - Profile information (name, email, contact)
 - Quick statistics (total simulations accessed, learning streak)
 - Recent activity feed

- Quick access links to simulations and help desk
- Notifications for events or updates
- **Outputs:** Rendered dashboard with real-time data
- **Priority:** High

FR-008: Profile Management

- **Description:** The system shall allow players to view and update their profile information
- **Editable Fields:** Name, contact number, password (with current password verification)
- **Non-Editable Fields:** Email address, account creation date
- **Process:** Update form with validation, save changes to database, confirmation message
- **Priority:** Medium

FR-009: Simulation Categories Navigation

- **Description:** The system shall present cricket techniques organized in hierarchical categories
- **Category Structure:**
 - Level 1: Batting / Bowling
 - Level 2 (Batting): Defensive Shots / Attacking Shots
 - Level 2 (Bowling): Fast Bowling / Spin Bowling / Medium-Pace Bowling
 - Level 3: Specific techniques (e.g., Cover Drive, Yorker)
- **Interface:** Card-based or menu-based navigation with icons and descriptions
- **Priority:** High

FR-010: Simulation Viewing

- **Description:** The system shall display interactive simulations with step-by-step visual guidance
- **Components:**
 - Visual animation/image sequence showing technique
 - Step-by-step textual description
 - Key points highlighting critical aspects
 - Common mistakes section

- Playback controls (play, pause, replay, adjust speed)
- **Inputs:** User selection of simulation, playback controls
- **Outputs:** Rendered simulation with synchronized text
- **Priority:** High

FR-011: Performance Tracking

- **Description:** The system shall automatically track and store player simulation access and usage patterns
- **Tracked Metrics:**
 - Simulations accessed (technique name, timestamp)
 - Time spent per simulation
 - Total time spent on platform
 - Techniques completed (viewed fully)
 - Learning streaks (consecutive days of activity)
- **Storage:** Data stored in user profile database
- **Priority:** High

FR-012: Statistics Visualization

- **Description:** The system shall display player performance statistics through charts and summaries
- **Visualizations:**
 - Progress charts (techniques learned over time)
 - Category breakdown (batting vs bowling time distribution)
 - Activity calendar showing daily usage
 - Milestone achievements (e.g., "10 techniques completed")
- **Priority:** Medium

FR-013: Help Desk Ticket Submission

- **Description:** The system shall allow players to submit queries or technical issues through help desk
- **Inputs:** Subject, category (Technical Issue / Content Question / Account Issue), description, optional attachment
- **Process:**

- (a) Player accesses Help Desk section
- (b) Player fills ticket submission form
- (c) System validates and saves ticket with "Open" status
- (d) System assigns unique ticket ID
- (e) Admin receives notification of new ticket
- (f) Player receives confirmation with ticket ID
- **Outputs:** Ticket created, notifications sent
- **Priority:** Medium

FR-014: Ticket Status Tracking

- **Description:** The system shall allow players to view status and history of submitted tickets
- **Display:** List of tickets with status (Open / In Progress / Resolved / Closed)
- **Details:** Ticket ID, subject, submission date, last update, admin response
- **Priority:** Low

FR-015: Event Viewing

- **Description:** The system shall display cricket-related events and announcements to players
- **Display:** List of upcoming and past events with details
- **Information:** Event title, date, venue, description, registration link (if applicable)
- **Priority:** Low

3.1.3 Admin Module Requirements

FR-016: Admin Dashboard

- **Description:** The system shall provide comprehensive admin dashboard with system overview
- **Components:**
 - User statistics (total users, pending approvals, active users)
 - Recent activity log
 - Open help desk tickets count
 - System health indicators
 - Quick action buttons
- **Priority:** High

FR-017: User List Management

- **Description:** The system shall provide admin interface to view and manage all user accounts
- **Features:**
 - Paginated user list with search functionality
 - Filter by account status (Pending / Active / Suspended)
 - Sort by registration date, name, activity
 - View detailed user profile and statistics
 - Actions: Approve, Suspend, Delete account
- **Priority:** High

FR-018: User Profile Viewing

- **Description:** The system shall allow admin to view complete user profiles and performance data
- **Viewable Data:**
 - Personal information (name, email, contact)
 - Account status and creation date
 - Complete activity history
 - Performance metrics and statistics
 - Submitted help desk tickets
- **Priority:** Medium

FR-019: Simulation Content Upload

- **Description:** The system shall allow admin to upload new simulation content
- **Inputs:**
 - Technique name and category
 - Visual assets (images/animation files)
 - Step-by-step textual instructions
 - Key points and common mistakes
 - Metadata (difficulty level, estimated time)
- **Process:** Form-based upload with file validation, content preview, publish action
- **Priority:** High

FR-020: Simulation Content Editing

- **Description:** The system shall allow admin to modify existing simulation content
- **Features:**
 - Edit text descriptions
 - Replace visual assets
 - Update metadata
 - Version history tracking
- **Priority:** Medium

FR-021: Help Desk Management

- **Description:** The system shall provide admin interface to manage and respond to user tickets
- **Features:**
 - View all tickets with filtering (Open / In Progress / Resolved / Closed)
 - Sort by submission date, priority, category
 - View ticket details and user information
 - Update ticket status
 - Add admin response/comments
 - Close resolved tickets
- **Notifications:** User receives email when admin responds or updates status
- **Priority:** Medium

FR-022: Event Management

- **Description:** The system shall allow admin to create, edit, and delete event announcements
- **Inputs:** Event title, date, time, venue, description, registration link, image (optional)
- **Operations:** Create new event, edit existing event, delete event, mark as past/archived
- **Display:** Events shown to players in chronological order
- **Priority:** Low

FR-023: Analytics and Reporting

- **Description:** The system shall generate usage analytics and reports for administrators
- **Reports:**
 - User growth trends
 - Most accessed simulations
 - Average time spent per user
 - Category popularity (batting vs bowling)
 - User engagement metrics
 - Help desk ticket trends
- **Export:** Reports exportable as CSV or PDF
- **Priority:** Low

3.2. External Interface Requirements

3.2.1 User Interface Requirements

UI-001: General UI Principles

- **Design Philosophy:** Clean, intuitive, cricket-themed interface
- **Color Scheme:** Cricket-inspired palette (greens, whites, blues) with high contrast for readability
- **Typography:** Clear, readable fonts; minimum 14pt for body text
- **Responsiveness:** Adaptive layout supporting desktop (1920x1080 to 1366x768) and mobile (375x667 to 414x896)
- **Navigation:** Consistent header/sidebar navigation across all pages
- **Feedback:** Visual feedback for all user actions (loading indicators, success/error messages)

UI-002: Player Interface Layout

- **Header:** Logo, navigation menu (Dashboard, Simulations, Help, Profile), logout button
- **Dashboard:** Grid/card layout for statistics and quick links
- **Simulation Browser:** Category cards with icons and technique counts
- **Simulation Viewer:** Full-width player area with controls and side panel for instructions
- **Forms:** Clear labels, inline validation, helpful error messages

UI-003: Admin Interface Layout

- **Sidebar:** Vertical navigation menu with sections (Dashboard, Users, Content, Help Desk, Events, Reports)
- **Data Tables:** Sortable, filterable tables with pagination and action buttons
- **Forms:** Comprehensive forms with rich text editors for content management
- **Modals:** Confirmation dialogs for destructive actions

3.2.2 Hardware Interface Requirements**HW-001: Client Device Requirements**

- **Minimum Specifications:**
 - Processor: 1 GHz dual-core
 - RAM: 2 GB
 - Display: 1024x768 resolution
 - Internet: 2 Mbps connection
 - Browser: Modern browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- **Recommended Specifications:**
 - Processor: 2 GHz quad-core
 - RAM: 4 GB+
 - Display: 1920x1080 resolution
 - Internet: 5 Mbps+ connection

HW-002: Server Requirements

- **Web Server:**
 - Cloud-based virtual machine or container
 - 2 CPU cores, 4 GB RAM minimum
 - 50 GB SSD storage for application
 - Auto-scaling capability for traffic spikes
- **Database Server:**
 - Managed database service or dedicated instance
 - 2 GB RAM minimum
 - 20 GB storage initially (scalable)

4. System Features

This section describes the major features of Namal Criko, organized by functional area. Each feature includes detailed description, stimulus/response sequences, and associated functional requirements.

4.1. Cricket Technique Simulation Engine

4.1.1 Description and Priority

The simulation engine is the core feature of Namal Criko, providing interactive visual demonstrations of cricket techniques. This is the highest priority feature as it delivers the primary value proposition.

Priority: High

4.1.2 Stimulus/Response Sequences

1. **Stimulus:** Player selects "Batting" category from dashboard
2. **Response:** System displays subcategories (Defensive, Attacking Shots)
3. **Stimulus:** Player selects "Attacking Shots"
4. **Response:** System displays list of attacking shot techniques
5. **Stimulus:** Player clicks on "Cover Drive"
6. **Response:** System loads and displays Cover Drive simulation with controls and instructions
7. **Stimulus:** Player clicks "Play" button
8. **Response:** Animation plays with synchronized textual steps
9. **Stimulus:** Player clicks "Replay" or adjusts speed
10. **Response:** System restarts animation or changes playback speed

4.1.3 Functional Requirements

- FR-009: Simulation Categories Navigation
- FR-010: Simulation Viewing
- FR-011: Performance Tracking (automatically log access)

4.2. User Account Management System

4.2.1 Description and Priority

Comprehensive system for user registration, authentication, profile management, and administrative approval workflow.

Priority: High

4.2.2 Stimulus/Response Sequences

1. **Stimulus:** New user visits platform and clicks "Register"
2. **Response:** System displays registration form
3. **Stimulus:** User submits completed form
4. **Response:** System validates, creates account with "Pending" status, sends confirmation email
5. **Stimulus:** Admin receives notification and logs into admin panel
6. **Response:** System displays pending approvals list
7. **Stimulus:** Admin reviews user details and clicks "Approve"
8. **Response:** System changes status to "Active", sends approval email to user
9. **Stimulus:** User attempts to login with credentials
10. **Response:** System authenticates and redirects to player dashboard

4.2.3 Functional Requirements

- FR-001: User Registration
- FR-002: Admin Account Approval
- FR-003: User Login (Player)
- FR-004: Admin Login
- FR-005: Password Reset
- FR-006: Session Management
- FR-008: Profile Management

4.3. Performance Analytics and Progress Tracking

4.3.1 Description and Priority

Automated tracking and visualization of player learning progress and usage patterns.

Priority: Medium

4.3.2 Stimulus/Response Sequences

1. **Stimulus:** Player accesses simulation
2. **Response:** System silently logs access timestamp, technique name, player ID
3. **Stimulus:** Player completes viewing simulation
4. **Response:** System calculates time spent, marks technique as "viewed"
5. **Stimulus:** Player navigates to "My Statistics" page
6. **Response:** System queries database, calculates metrics, generates visualizations
7. **Stimulus:** Player views progress chart
8. **Response:** System displays line chart of techniques learned over time

4.3.3 Functional Requirements

- FR-011: Performance Tracking
- FR-012: Statistics Visualization

4.4. Help Desk Ticketing System

4.4.1 Description and Priority

Support system enabling players to submit queries and admins to manage/respond to tickets.

Priority: Medium

4.4.2 Stimulus/Response Sequences

1. **Stimulus:** Player encounters issue and accesses Help Desk
2. **Response:** System displays ticket submission form
3. **Stimulus:** Player fills form with issue details and submits
4. **Response:** System creates ticket with unique ID, notifies admin, shows confirmation

5. **Stimulus:** Admin logs in and views Help Desk panel
6. **Response:** System displays list of open tickets
7. **Stimulus:** Admin selects ticket and adds response
8. **Response:** System saves response, sends email to player, updates ticket status
9. **Stimulus:** Admin marks ticket as "Resolved"
10. **Response:** System updates status, notifies player

4.4.3 Functional Requirements

- FR-013: Help Desk Ticket Submission
- FR-014: Ticket Status Tracking
- FR-021: Help Desk Management

4.5. Content Management System

4.5.1 Description and Priority

Administrative interface for uploading, editing, and managing simulation content.

Priority: High

4.5.2 Stimulus/Response Sequences

1. **Stimulus:** Admin decides to add new bowling technique
2. **Response:** Admin navigates to Content Management section
3. **Stimulus:** Admin clicks "Add New Simulation"
4. **Response:** System displays content upload form
5. **Stimulus:** Admin fills form with technique details, uploads assets
6. **Response:** System validates files, saves to database and file storage
7. **Stimulus:** Admin clicks "Publish"
8. **Response:** System makes content visible to players, shows success confirmation

4.5.3 Functional Requirements

- FR-019: Simulation Content Upload
- FR-020: Simulation Content Editing

4.6. Event and Announcement Management

4.6.1 Description and Priority

Feature allowing admins to post cricket-related events and players to view them.

Priority: Low

4.6.2 Stimulus/Response Sequences

1. **Stimulus:** Admin wants to announce upcoming cricket match
2. **Response:** Admin accesses Event Management section
3. **Stimulus:** Admin fills event details form and submits
4. **Response:** System saves event, displays in events list for players
5. **Stimulus:** Player navigates to Events page
6. **Response:** System displays chronological list of upcoming and past events

4.6.3 Functional Requirements

- FR-015: Event Viewing
- FR-022: Event Management

5. Non-Functional Requirements

5.1. Performance Requirements

NFR-001: Response Time

- Page load time shall not exceed 3 seconds on standard broadband (5 Mbps)
- Simulation playback shall start within 2 seconds of user clicking "Play"
- Database queries shall return results within 1 second for 95% of requests
- Search functionality shall display results within 2 seconds

NFR-002: Concurrent Users

- System shall support minimum 500 concurrent users without significant degradation
- System shall scale to handle 1000 concurrent users with optimized performance
- Database shall handle 100 simultaneous read/write operations

NFR-003: Throughput

- Web server shall handle 50 requests per second during peak usage
- Static assets (images, CSS, JS) shall be delivered efficiently via caching
- API endpoints shall process 30 requests per second for dynamic content

NFR-004: Resource Utilization

- Server CPU usage shall not exceed 80% under normal load
- Memory usage shall remain below 70% of allocated capacity
- Database storage shall be optimized with indexing for frequently queried fields

5.2. Safety Requirements

NFR-005: Data Safety

- System shall implement regular automated database backups (daily incremental, weekly full)
- Backup retention period shall be minimum 30 days
- Disaster recovery plan shall ensure system restoration within 24 hours

NFR-006: User Data Protection

- Personal user information (passwords) shall be encrypted using industry-standard hashing algorithms (bcrypt)
- Passwords shall never be stored in plain text or logged
- Password reset tokens shall be single-use and expire after 1 hour

5.3. Security Requirements**NFR-007: Authentication Security**

- All authentication attempts shall be logged for security audit
- Accounts shall be locked after 5 consecutive failed login attempts
- Session tokens shall be invalidated after 120 minutes of inactivity
- Re-authentication required for sensitive operations (password change, email update)

NFR-008: Authorization Controls

- Role-based access control (RBAC) shall enforce separation between player and admin functions
- Admin actions shall require elevated privilege verification
- Users shall only access their own data unless specifically authorized
- API endpoints shall validate user permissions for each request

NFR-009: Data Transmission Security

- All client-server communication shall use HTTPS (TLS 1.2 or higher)
- Secure headers (HSTS, CSP) shall be implemented to prevent common web vulnerabilities
- Cookies shall be marked as secure and HTTP-only where applicable

NFR-010: Input Validation

- All user inputs shall be validated both client-side and server-side
- SQL injection prevention through parameterized queries or ORM
- Cross-site scripting (XSS) protection through output encoding
- File uploads shall be scanned for malware and restricted to approved formats

5.4. Software Quality Attributes

NFR-011: Reliability

- System shall maintain 99% uptime during academic periods
- Mean Time Between Failures (MTBF) shall exceed 720 hours (30 days)
- Mean Time To Repair (MTTR) shall be less than 4 hours for critical failures
- Error rate shall not exceed 1% of all user interactions

NFR-012: Availability

- System shall be available 24/7 with scheduled maintenance windows announced in advance
- Scheduled maintenance shall not exceed 4 hours per month
- High-availability architecture with redundant components where feasible
- Graceful degradation when dependent services (email) are unavailable

NFR-013: Maintainability

- Codebase shall follow consistent coding standards and documentation
- Modular architecture shall allow independent updates to components
- Configuration shall be externalized for easy environment-specific adjustments
- Comprehensive logging shall facilitate debugging and issue resolution

NFR-014: Scalability

- Architecture shall support horizontal scaling for increased user load
- Database design shall accommodate growth to 10,000 users
- Simulation content shall scale to 500+ techniques without performance degradation
- Stateless design shall allow load distribution across multiple servers

NFR-015: Usability

- System shall achieve 85% satisfaction rating in user acceptance testing
- New users shall be able to complete first simulation within 5 minutes of registration
- Interface shall comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA
- Help documentation shall be context-sensitive and easily accessible

NFR-016: Portability

- Web application shall function correctly across Chrome, Firefox, Safari, and Edge browsers
- Responsive design shall provide optimal experience on desktop, tablet, and mobile devices
- Application shall be deployable on major cloud platforms (AWS, Azure, GCP)
- Database migration shall be scripted and reversible

5.5. Operational Requirements**NFR-017: Deployment Requirements**

- System shall support automated deployment using CI/CD pipelines
- Database migrations shall be version-controlled and executable without downtime
- Rollback procedure shall be documented and tested for failed deployments
- Environment configuration shall be separate from application code

NFR-018: Monitoring Requirements

- System shall log all errors and critical events for monitoring
- Performance metrics (response time, error rates, user counts) shall be collected
- Alerts shall be configured for system failures or performance degradation
- Admin dashboard shall display key system health indicators

NFR-019: Backup Requirements

- Full system backup shall be performed weekly
- Incremental backups shall be performed daily
- Backup restoration procedure shall be tested quarterly
- Backups shall be stored securely with encryption

NFR-020: Compliance Requirements

- System shall comply with Namal University IT policies
- Data handling shall follow university privacy guidelines
- All third-party components shall be properly licensed for educational use
- Security audit shall be conducted before production deployment

6. Other Requirements

6.1. Development Requirements

OR-001: Development Methodology

- Project shall follow Agile development methodology with 2-week sprints
- Regular code reviews shall be conducted for quality assurance
- Version control shall be maintained using Git with descriptive commit messages
- Continuous integration shall be implemented with automated testing

OR-002: Documentation Requirements

- Comprehensive technical documentation shall be maintained
- API documentation shall be generated and kept current
- User manuals shall be created for both player and admin roles
- Deployment and maintenance procedures shall be documented

OR-003: Testing Requirements

- Unit testing shall achieve 80% code coverage for critical modules
- Integration testing shall validate all system components work together
- User acceptance testing shall be conducted with representative users
- Performance testing shall validate system under expected load conditions
- Security testing shall include vulnerability scanning and penetration testing

OR-004: Coding Standards

- Code shall follow consistent naming conventions and formatting
- Comments shall explain complex algorithms and business logic
- Error handling shall be implemented for all possible failure scenarios
- Configuration shall be separated from code for different environments

6.2. Legal and Regulatory Requirements

OR-005: Intellectual Property

- All source code shall be owned by Namal University
- Third-party libraries shall be properly attributed and licensed
- Cricket technique content shall be original or properly licensed
- User agreement shall specify terms of use and intellectual property rights

OR-006: Data Privacy

- Privacy policy shall comply with applicable data protection regulations
- User consent shall be obtained for data collection and processing
- Data retention periods shall be defined and implemented
- User data export and deletion procedures shall be available

OR-007: Accessibility Compliance

- System shall comply with Web Content Accessibility Guidelines (WCAG) 2.1 Level AA
- Accessibility testing shall be conducted with screen readers
- Keyboard navigation shall be fully supported
- Alternative text shall be provided for all non-text content

6.3. System Support Requirements

OR-008: Training Requirements

- Administrator training shall be provided for system management
- User orientation materials shall be developed for new players
- Technical documentation shall be maintained for development team

OR-009: Maintenance Requirements

- Regular security updates shall be applied to all system components
- Performance monitoring shall be conducted continuously
- Backup procedures shall be tested quarterly
- Issue tracking system shall be maintained for bug reports

OR-010: Disaster Recovery

- Disaster recovery plan shall be documented and tested annually
- Critical system recovery time objective (RTO) shall be 24 hours
- Data recovery point objective (RPO) shall be 24 hours
- Backup systems shall be located separately from primary systems

6.4. Sustainability Requirements**OR-011: Environmental Considerations**

- System shall optimize resource usage to minimize energy consumption
- Code efficiency shall be prioritized to reduce server load
- Sustainable hosting options shall be considered where available

OR-012: Long-term Viability

- System architecture shall support future enhancements
- Technology stack shall use well-supported frameworks and libraries
- Documentation shall enable knowledge transfer to new team members
- Exit strategy shall be defined for system decommissioning if needed

A. Appendix A: Glossary

Term	Detailed Definition
Simulation	Interactive digital representation of cricket technique showing body movements, timing, and execution steps through visual animations accompanied by textual instructions.
Player	Registered user who accesses the platform to learn cricket techniques through simulations. Players have limited permissions focused on learning activities.
Administrator	Privileged user responsible for managing the platform, including user approvals, content management, help desk resolution, and system monitoring.

Term	Detailed Definition
Help Desk Ticket	Formal record of user query or issue submitted through the platform's help desk system, tracked through resolution lifecycle.
Performance Tracking	Automated system for recording user interactions with simulations, including access frequency, duration, and completion metrics.
Account Approval Workflow	Process where new user registrations require manual review and approval by administrator before granting system access.
Content Management	Administrative functions for uploading, editing, organizing, and publishing simulation content and platform announcements.
User Dashboard	Personalized interface displaying user-specific information, statistics, quick links, and notifications.
Admin Dashboard	Management interface providing system overview, user statistics, content controls, and administrative tools.
Session Management	System mechanism for maintaining authenticated user state across multiple requests with timeout and security controls.
Event Management	Feature allowing administrators to post cricket-related events, tournaments, or announcements for user visibility.
Responsive Design	Web design approach ensuring optimal viewing and interaction experience across range of devices from desktop to mobile.
CRUD Operations	Fundamental data operations: Create (add new records), Read (retrieve records), Update (modify records), Delete (remove records).
API Endpoint	Specific URL where API can be accessed to perform certain operation or retrieve specific data.
Authentication	Process of verifying user identity through credentials (email/password).
Authorization	Process of determining what authenticated user is allowed to access or perform within system.
Data Encryption	Conversion of data into coded form to prevent unauthorized access during transmission or storage.
Load Testing	Performance testing to determine system behavior under expected and peak user load conditions.

Term	Detailed Definition
Uptime	Measure of time system is operational and available to users, expressed as percentage of total time.
Version Control	System for tracking changes to source code over time, enabling collaboration and change management.

B. Appendix B: Analysis Models

B.1. Use Case Diagram

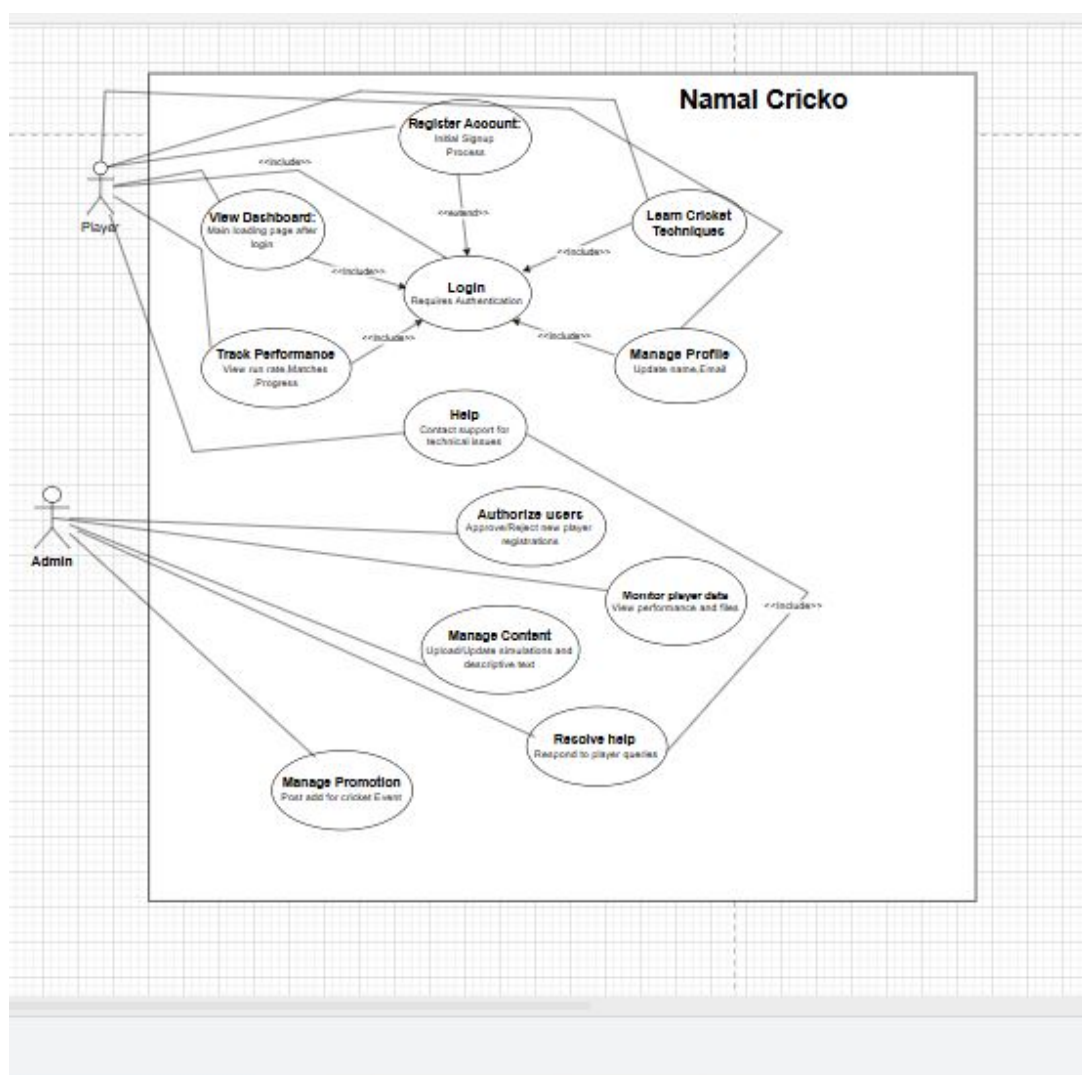


Figure 2: High-level Use Case Diagram showing actors (Player, Admin) and major system interactions

C. Appendix C: To Be Determined List

- **TBD-001:** Final selection of technology stack (frameworks, libraries)
- **TBD-002:** Hosting platform selection (university servers vs cloud)
- **TBD-003:** Database system selection (MySQL vs PostgreSQL vs MongoDB)
- **TBD-004:** Email service provider for system notifications
- **TBD-005:** Specific cricket techniques to be included in initial release
- **TBD-006:** Visual assets source (custom creation vs licensed content)
- **TBD-007:** Domain name registration and configuration
- **TBD-008:** SSL certificate provider and configuration
- **TBD-009:** Backup storage location and retention policy details
- **TBD-010:** Monitoring tool selection and configuration
- **TBD-011:** User acceptance testing participant selection
- **TBD-012:** Go-live date and deployment schedule